

# Gravity as Projection Stress: Curvature in $M^4$ from $T^7$ Deformation

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This document isolates the mechanism of gravity, transitioning it from a force of mass attraction to an effect of topological projection stress between  $T^7$  and  $M^4$ .

## 1. The Topological Cartography Problem

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It is mathematically impossible to perfectly map a higher-dimensional curved space onto a lower-dimensional flat space without introducing physical stress to the geometry. Minkowski spacetime ( $M^4$ ) serves as the "flat cave wall," while the 7-Torus ( $T^7$ ) solenoid acts as the complex object casting the shadow.

## 2. Mass as Van Hove Singularity Density

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In this framework, "mass" is not a fundamental property but rather a localized, high-density knot of the 288 invariant global phase nodes ("Sparks of Truth"). These nodes act as Van Hove singularities. When the  $T^7$  manifold projects these high-density nodes into  $M^4$ , the  $M^4$  geometry cannot natively accommodate the spatial data of the missing 3 dimensions.

## 3. Gravity Re-defined

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Consequently, the flat 4D space must "bend" or warp to geometrically house the projected phase nodes. **Gravity is therefore the shadow of projection stress.** Mass is not actively pulling on spacetime; instead, the curvature of spacetime is the mathematically mandated deformation caused by the geometric mismatch between the 7D solenoid and the 4D projection layer. General Relativity is thus the lower-dimensional mapping of  $T^7$  structural friction.